

Embedding-Based Distribution-Shift Metric — Prototype Pipeline

Goal: build and validate a first working "images → embeddings → distribution-shift score" pipeline, as a prototype step ahead of the full 10-configuration experiment.

Datasets involved: Cityscapes (real), Synscapes (synthetic, smaller reality gap), GTA5 (synthetic, larger reality gap).

Step 1 — Choose an embedding model

Pick **one** of three candidate backbones and build the pipeline around it first (not all three at once):

Model	Role	Notes
<code>facebook/dinov2-base</code>	General-purpose, self-supervised ViT	768-d, strong urban transfer, no confounds, has FID/KID literature precedent (FD-DINOv2). Recommended starting point.
<code>geolocal/StreetCLIP</code>	CLIP ViT-L/14 fine-tuned on 1.1M street images	768-d, true street-pretrained embedder, but trained for geolocalization → may confound geographic style (e.g. "looks American") with segmentation-relevant structural shift.
<code>facebook/dpt-dinov2-large-kitti</code>	DINOv2-large backbone, fine-tuned for depth on KITTI	1024-d, must discard the DPT depth head and keep only backbone features. Sensitive to scene geometry/layout.

All three are **embedding-producing backbones**, not segmentation models — segmentation training is a separate, later part of the project.

Step 2 — Extract embeddings

Run the chosen model (e.g. DINOv2-base) over each dataset sample to turn images into fixed-length vectors:

1. Load pretrained model + its image processor.
2. Load image samples from each dataset/subset folder.
3. Preprocess (resize, normalize, tensor conversion — handled by the model's own processor).

4. Process images in batches.
5. Extract one embedding per image — **CLS token** from `(last_hidden_state)` (matches FD-DINOv2 precedent; document this choice).
6. Do **not** L2-normalize (the Fréchet distance in Step 3 needs raw statistics).
7. Save each dataset's embeddings as `(.npz)` — shape `(N_sample, 768)`.

Run this once per dataset/subset needed for Step 4's checks:

- Cityscapes subset A
- Cityscapes subset B
- Synscapes
- GTA5

Note: this uses **sample subsets** (e.g. ~500 images each), not full datasets — sufficient for prototype validation, not the final full-scale run.

Step 3 — Compute distance metrics

Each dataset's pooled embeddings form **one distribution**, characterized by:

- μ — mean vector (centroid)
- Σ — covariance (spread/shape)

Two metrics, in order of implementation:

1. **Centroid distance** (simplest): $d(A, B) = ||\mu_A - \mu_B||_2$ — compares only where the two embedding clouds are centered.
2. **Fréchet-style distance**: $d^2 = ||\mu_A - \mu_B||^2 + \text{Tr}(\Sigma_A + \Sigma_B - 2(\Sigma_A \Sigma_B)^{\frac{1}{2}})$ — also accounts for spread/shape. Same math as FID, just with DINOv2 features (i.e. a domain-adapted FID-analog, not a "CLIP distance" analog).

With 3 real dataset pairs (Cityscapes-Synscapes, Cityscapes-GTA5, Synscapes-GTA5) plus the Cityscapes-A vs. Cityscapes-B self-check:

- **4 distance values** using centroid distance only
 - **8 distance values** if both metrics are computed
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Step 4 — Sanity checks

Verify the pipeline behaves as expected before trusting any numbers:

Comparison	Expected result	Purpose
Cityscapes-A vs. Cityscapes-B	≈ 0	Core correctness check
Cityscapes vs. Synscapes	Small	Smaller reality gap
Cityscapes vs. GTA5	Larger than Cityscapes–Synscapes	Larger reality gap
Synscapes vs. GTA5	—	Optional/bonus context

A plausible-looking number alone doesn't prove correctness — these checks are a gate: pass them and the pipeline is ready to scale up; fail them and there's a bug (or DINOv2's embedding space doesn't track the reality gap as expected) to resolve first.

Step 5 — Document practical decisions

Record every choice that affects the resulting distance number, so results are reproducible and comparable later:

- Selected model + exact checkpoint
- Embedding dimension
- Model output used (CLS token vs. mean-pooled patches)
- Whether embeddings were normalized
- Preprocessing details
- Pooling method
- Number of images sampled per dataset
- Batch size
- Runtime / memory footprint

This is not a separate technical task — it's the paper trail that makes Steps 1-4 reproducible, comparable across the three candidate models later, and defensible when reporting results (including the eventual paper).

Scope of this prototype

In scope: one model, one dataset pair (minimum), one metric, sanity checks, documentation.

Not yet in scope: all 10 hybrid configurations, all three candidate models, segmentation model training, mIoU correlation, final metric selection.